gpeyt1-46 · Driven pile in clay (α)

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

Problem

A 16 in. square precast concrete pile is driven 50 ft into a two-layer clay deposit: 0–20 ft soft clay su = 600 psf (α = 0.95); 20–60 ft stiff clay su = 1400 psf (α = 0.62). The water table is at 5 ft. Clay γsat = 115 pcf. Use the α-method for side resistance and qp = 9su at the toe. Design load 90 kips; ASD FS = 2.5.

Most nearly, the ultimate axial compression capacity Ru is:

A. 310 kips
B. 200 kips
C. 174 kips
D. 222 kips

Given equations / data

> Rs = Σ α su (perimeter)(ΔL); Rt = 9 su At; Ru = Rs + Rt

Worked solution

1. Perimeter = 4×1.333 = 5.333 ft; At = 1.778 ft²
2. Layer 1: 0.95×600×5.333×20 = 60,800 lb
3. Layer 2: 0.62×1400×5.333×30 = 138,880 lb
4. Rt = 9×1400×1.778 = 22,400 lb
5. Ru = 222,080 lb ≈ 222 kips (Rall = 89 kips at FS 2.5 > 90 ✓)

Final answer

222 kips

Why the other values are wrong

- **310 kips** — omitted α (α = 1)
- **200 kips** — left out toe resistance
- **174 kips** — treated the pile as a 16 in. round pile

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) Vol I §7.2 α-method (Eq. 7-31) and qp = 9su (Eq. 7-16)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=hfbcwIJfAjw, https://www.youtube.com/watch?v=Jdcs8-i_OA4, https://www.youtube.com/watch?v=ZcjQbEB66IQ (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.